

Roll No.

TEE-501

B. TECH. (EE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

POWER SYSTEMS-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Compare the volume of conductor material required for 3-phase ac three wire and DC two wire system with mid-point earth. (CO1)

OR

- (b) A single-phase ac system supplies a load of 200 kW and if this system is converted to 3-phase, 3-wire system by running a third similar conductor, calculate the 3-phase load that can now be supplied if the voltage between the conductors is the same. Assume the power factor and transmission efficiency to be same in the two cases. (CO1)
2. (a) Derive the relation between string efficiency and voltage distributions across the disc of suspension type insulators. Assume three discs in the systems. (CO1)

P. T. O.

OR

- (b) An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency. (CO1)
3. (a) Discuss the phenomenon of corona existing in high voltage transmission systems. (CO1)

OR

- (b) A certain 3-phase equilateral transmission line has a total corona loss of 53 kW at 160 kV and a loss of 98 kW at 110.9 kV. What is disruptive critical voltage ? What is the corona loss at 113 kV ? (CO1)
4. (a) Derive the expression of mechanical sag for the unequal support. Also, discuss the effects of wind pressure on mechanical sag. (CO2)

OR

- (b) A transmission line has a span of 300 m between level supports. The conductor has an effective diameter of 2 cm and weighs 0.965 kg/m. Its ultimate strength is 8060 kg. If the conductor has ice coating of radial thickness 1.27 cm and is subjected to a wind pressure of 3.9 gm/cm^2 of the projected area. Calculate sag for a safety factor of 2. Weight of 1 c.c. of ice is 0.91 gm. (CO3)

(3)

5. (a) Discuss why high transmission level voltage has been used for transmission of electrical power. (CO2)

OR

- (b) Derive the inductance of three-phase three-wire transmission line. Assume symmetrical spacing between the conductors. (CO3)

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CONTROL SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Compare the open loop and closed loop control system.
(ii) What do you mean by type and order of the system ?
(iii) What will be the steady state error of the type 1 system when subjected to a unit parabolic input ?
(iv) Write the relation between the unit step and unit ramp signal.
(v) The open loop transfer function of a unity feedback control system is : (CO1/CO2)

$$G(s) = \frac{1}{(s + 2)^2}$$

Determine the closed loop transfer poles.

P. T. O.

(2)

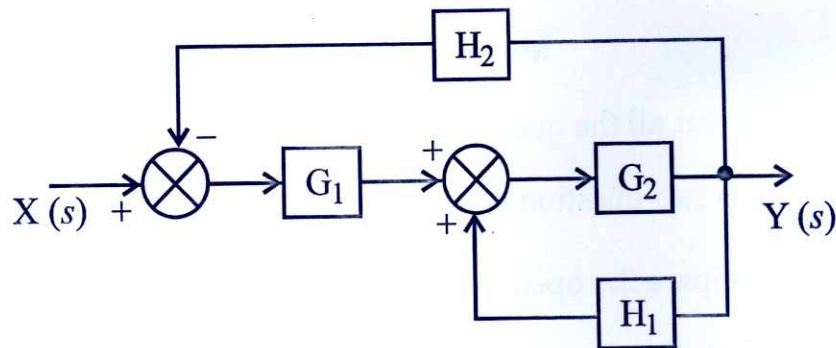
TEE-502

OR

- (b) (i) What is Mason's gain formula ?
(ii) What is Impulse response of the system ?
(iii) Define Damping ratio of the control system.
(iv) What is steady state error of the control system ?
(v) What is the setting time for the second order control system ?

(CO1/CO2)

2. (a) Determine the transfer function of the given system : (CO1/CO2)



OR

- (b) Derive the sensitivity of an overall transfer function of a closed loop control system with respect to the forward path transfer function $G(s)$.

(CO1/CO2)

3. (a) What is the open loop transfer function for the system whose characteristic equation is : (CO1/CO2)

$$F(s) = s^3 + 3s^2 + (k + 2)s + 5k = 0$$

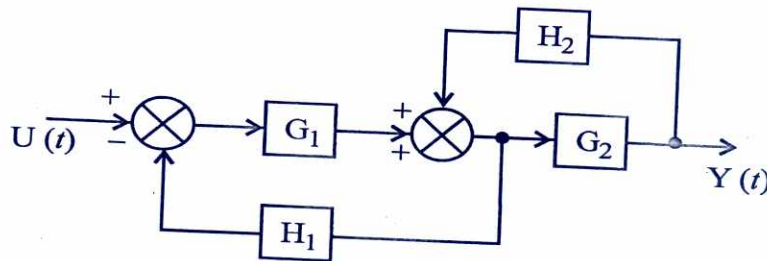
(3)

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OR

(b) Find out the transfer function of the given system :

(CO1/CO2)



4. (a) A second order control system is defined by the following differential equation :
(CO1/CO2)

$$4 \frac{d^2 c(t)}{dt^2} + 8 \frac{dc(t)}{dt} + 16c(t) = 16u(t)$$

Determine the damping ratio and natural frequency of the above system.

OR

- (b) The open loop transfer function of a unity feedback system is given by :

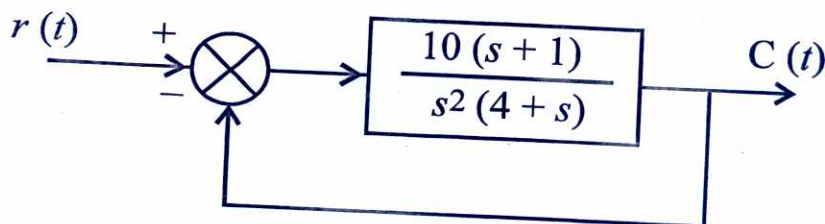
$$\frac{k}{s(s+1)}$$

If the value of gain K is such that the system is critically damped. Then find out the closed loop poles of the system.
(CO1/CO2)

5. (a) The Steady state error resulting from an input :
(CO1/CO2/CO3)

$$r(t) = 2 + 3t + 4t^2$$

For the given system is :



P. T. O.

(4)

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OR

- (b) The transient response of second order under damped system starting from rest is given by :

$$c(t) = Ac^{-6t} \sin(8t + \theta) : t \geq 0$$

Determine the settling time, peak time and peak overshoot of the system.

(CO1/CO2/CO3)

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TEE-503

**B. TECH. (EE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

MICROPROCESSORS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write a short note on evolution of microprocessor. (CO1)

OR

(b) Explain the function of the following pins of 8085 microprocessor :

(i) INTR

(ii) SID

(iii) CLK OUT

(iv) S1, S0

(v) RESET IN' (CO1)

2. (a) Explain the register organization of 8085 microprocessor. (CO1)

OR

(b) (i) How the various control signals are generated in 8085 microprocessor ? Explain.

(ii) Draw and explain the function of microprocessor based computer.

(CO1)

P. T. O.

(2)

3. (a) Draw the timing diagram of the following instruction : (CO2)

MVI M, 00

OR

- (b) What do you understand by interrupt in microprocessor ? Classify the interrupts available in 8085 microprocessor. (CO2)

4. (a) Define addressing mode in microprocessor. How many types of addressing modes are available in 8085 microprocessor ? Explain each type with the help of examples. (CO2)

OR

- (b) Write an assembly language program to find the largest number in an array in 8085 microprocessor. (CO2)

5. (a) How many machine cycles will be required in the execution of the following instructions ? Write their name also :

(i) MOV A, B

(ii) LDA 3000H

(iii) STAX B

(iv) LXI H 3050H

(v) LHLD 3001H

(CO2)

OR

- (b) Write an assembly language program to add two 16 bit numbers in 8085 microprocessor. (CO2)

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TEE-504

B. TECH. (EE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ELECTRICAL MACHINE DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Differentiate between a single-phase core type transformer and shell type transformer. (CO1)

OR

- (b) Discuss window space factor in transformer design of three phase shell type transformer. (CO1)
2. (a) Discuss the challenges for good design in details. (CO2)

OR

- (b) What are factors affecting in designing of electrical machines ? Discuss in detail. (CO2)
3. (a) Derive the relation for KVA rating of core type single phase transformer. (CO1)

P. T. O.

(2)

OR

(b) Derive the relation for KVA rating of three phase core type transformer.

(CO1)

4. (a) What are the factors to be considered for design of flux density of the transformer ?

(CO2)

OR

(b) Enumerate the importance of No-load current of transformer with suitable diagram.

(CO2)

5. (a) Deduce the equation for voltage per turn of transformer and explain the role of flux in it.

(CO3)

OR

(b) What are various properties of electrical material and magnetic material ? Discuss in detail.

(CO3)

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UCE-501

B. TECH. (EE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

DISASTER MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is Disaster Management ? Explain various phases of Disaster Management Cycle. (CO1)

OR

- (b) Define disaster. Describe natural disaster and man-made disasters with examples of each. (CO1)

2. (a) How does the Tsunami develop ? What are its effects ? (CO1/CO2)

OR

- (b) Define epidemics. Give example and how do you manage ? (CO1/CO2)

3. (a) What are the factors that affect people's vulnerability and explain it.

(CO2)

OR

- (b) Describe the role of IT in disaster management.

(CO2)

P. T. O.

(2)

4. (a) Enumerate the role of media in disaster management. (CO1/CO3)

OR

- (b) Why is Disaster Management crucial for India ? (CO1/CO3)

5. (a) Write a note on any one recent disaster that occur in India. (CO1/CO3)

OR

- (b) Explain how development activities are the cause of disaster with proper example. (CO1/CO3)

10/10/23
1st shift

Roll No.

TOE-502

B. TECH. (EE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) The error in the measurement of the area of a circle is not allowed to exceed 0.1%. How accurately should the diameter be measured ?

(CO1)

OR

- (b) Write a short note on errors in numerical computation. (CO1)

2. (a) Find the real root of the equation $x \log_{10} x = 1.2$ by Bisection method correct to four decimal places. Taking $a = 2.74$ and $b = 2.75$, perform three iterations. (CO1)

OR

- (b) Find the cube root of 15 correct to four significant figures by iterative method. (CO1)

P. T. O.

3. (a) Find the root of the equation $xe^x = \cos x$ in the interval $(0, 1)$ using Regula-Falsi method. Perform two iterations. (CO1)

OR

- (b) Find a real root of the equation $x = e^{-x}$ using the Newton-Raphson method. Taking x_0 as 1. (CO1)
4. (a) From the following table of half-yearly premium for policies maturing at different ages, estimate the premium for a policy maturing at the age 63 : (CO2)

Age	Premium (in dollars)
45	114.84
50	96.16
55	83.32
60	74.48
65	68.48

OR

- (b) From the following table of half-yearly premiums for policies maturing at different ages, estimate the premium for policies maturing at age of 46 : (CO2)

Age	Premium (in dollars)
45	114.84
50	96.16
55	83.32
60	74.48
65	68.48

(3)

5. (a) Discuss Interpolation. Differentiate between argument and entry. (CO2)

OR

(b) Using Lagrange's interpolation formula, find $y(10)$ from the following table : (CO2)

x	y
5	12
6	13
9	14
11	16